
GoldBarn

Brief Project Overview

Core Requirements

Real-time Streaming: Spot price ingestion for Gold and Silver.

Interactive Visualization: Dynamic multi-frame chart canvas (1W, 1M, 6M, 1Y, Lifetime) + ledger history table.

Customizable Controls: Adjustable polling refresh intervals and multi-currency portfolio conversion (AED, GBP, CAD, EUR).

Automated Trading Signals: Real-time BUY / SELL / HOLD execution badges based on automated boundary evaluation



Architecture

Java Backend Core: Handles continuous polling loops, data modeling, and SQLite file I/O via native drivers.

Micro-HTTP Server: Exposes a lightweight endpoint at `http://localhost:5000/pricesGands`.

Electron UI Shell: Renders a clean grid layout using async fetch calls, keeping the UI smooth and responsive.

AI Trading Model & Automated Retraining Loop .

- **Signal Logic:** Tracks price variances against statistical thresholds (`differential_limit_gold / silver`) to trigger trading badges.
- **Persistent Logging:** Every price tick is saved to `stored_rates.db` using the `datalink1` connection.
- **Automated Model Retraining:** When dataset milestones are met (tracked by sample size `number_of_golds`), the backend calculates the Mean Absolute Percentage Error (MAPE Variance).
- **Adaptive Boundaries:** Dynamically recalibrates thresholds to stabilize prediction accuracy as market trends shift over time.